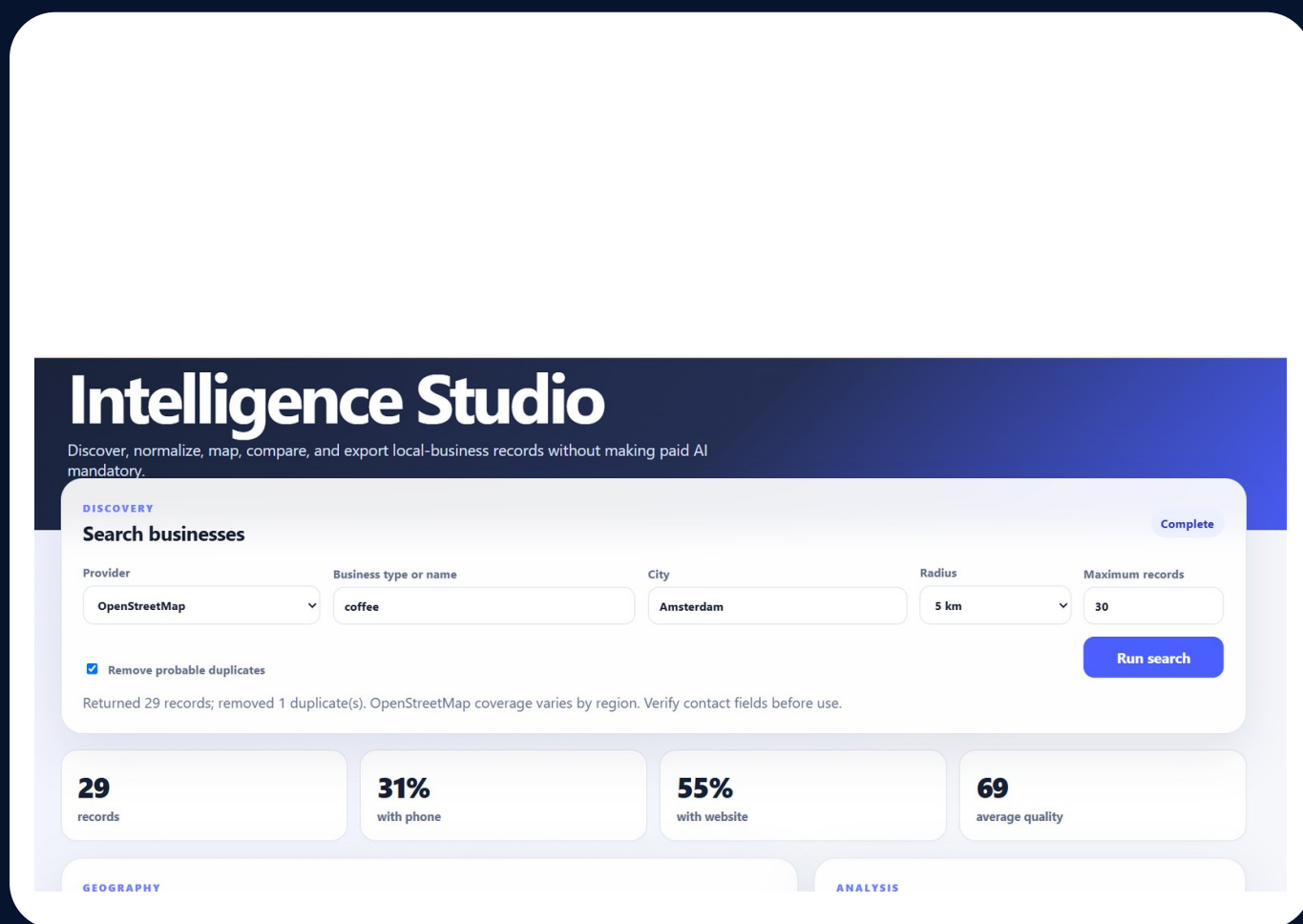


Inside GeoBusiness Intelligence Studio

A Visual Guide to Open Data, Geospatial Discovery, and Local Business Intelligence



Faramarz Kowsari

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Why build a local-business intelligence laboratory?

Business-location data is often fragmented, incomplete, duplicated, and difficult to compare.

OPEN

Accessible core

Sample data and OpenStreetMap keep the main workflow useful without purchasing an AI key.

CLEAR

Visible provenance

Every normalized record keeps its source visible so fictional, open, and commercial data remain distinguishable.

SAFE

Responsible boundaries

No Google Maps website scraping, CAPTCHA bypass, identity rotation, or access-control evasion.

USEFUL

Portfolio depth

FastAPI, adapters, validation, mapping, export, tests, Docker, CI, and optional AI in one coherent repository.

The project treats each provider as a source to be inspected, normalized, attributed, and verified — not as unquestionable truth.

One interface, three data paths

Sample provider

Fictional local records support offline demonstrations, predictable tests, and onboarding without network dependencies.

OpenStreetMap

Nominatim resolves the city and Overpass retrieves matching map objects. No key is required, but public-service policies matter.

Google Places

An optional adapter uses the official Places API Text Search endpoint with the user's own restricted key.

Future adapters

The shared provider contract allows municipal data, open directories, or authorized internal sources to be added later.

Provider output becomes useful only after it is converted into a shared business model with visible provenance and explicit limitations.

From a query to an exportable dataset

1

Choose provider, business type, city, radius, and limit.

2

Resolve geography and retrieve provider-specific records.

3

Normalize names, categories, addresses, contacts, ratings, and coordinates.

4

Score completeness and remove probable duplicates.

5

Map, summarize, inspect, and export CSV or JSON.

The deterministic path remains complete without AI. Model-generated interpretation is optional and never replaces source evidence.

A dashboard that keeps evidence visible

Intelligence Studio

Discover, normalize, map, compare, and export local-business records without making paid AI mandatory.

DISCOVERY

Search businesses

Complete

Provider

Business type or name

City

Radius

Maximum records

OpenStreetMap

coffee

Amsterdam

5 km

30

☒ Remove probable duplicates

Run search

Returned 29 records; removed 1 duplicate(s). OpenStreetMap coverage varies by region. Verify contact fields before use.

29

records

31%

with phone

55%

with website

69

average quality

GEOGRAPHY

ANALYSIS

Search controls

Provider, business type, city, radius, record limit, and duplicate removal remain explicit.

Coverage metrics

Record count, phone coverage, website coverage, and average completeness are visible at a glance.

Spatial evidence

The map keeps location and density visible instead of reducing the result to a spreadsheet.

Inspectable outputs

The table preserves provenance and quality; CSV and JSON support downstream analysis.

FastAPI at the center of replaceable components

Browser dashboard

Static HTML, CSS, and JavaScript served by FastAPI; Leaflet renders optional map layers.

API layer

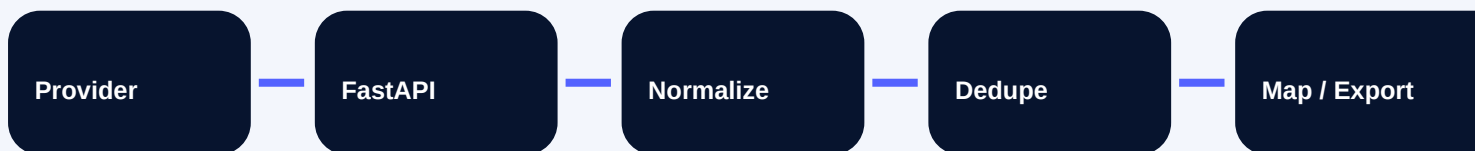
Pydantic validates requests and responses while OpenAPI exposes interactive endpoint documentation.

Search service

Provider selection, normalization, quality scoring, sorting, and probable duplicate removal.

Analysis layer

Built-in statistical briefing, plus optional Ollama or OpenAI-compatible chat completion.



Completeness is measured; identity is inferred cautiously

The project does not pretend that a single score proves correctness. The quality value reflects the presence of useful fields, while duplicate detection combines stronger evidence such as matching phone numbers with softer evidence such as similar names and addresses.

Normalization

Provider-specific payloads become a consistent business model with explicit source labels.

Completeness score

Phone, website, address, category, coordinates, and other fields contribute to an interpretable score.

Probable duplicates

Duplicates are removed conservatively; users can disable the feature and inspect raw provider output.

Verification warning

Contact fields and business status can change. Exported records must be checked before operational use.

Useful without AI; extensible when AI is justified

No paid key required

Discovery, normalization, mapping, deduplication, quality scoring, export, and deterministic analysis remain available without commercial AI.

Local option

Ollama can provide local inference without sending the dataset to a paid model provider.

BYOK integration

An OpenAI-compatible endpoint can be enabled with the user's own secret in a local `.env` file.

Secret hygiene

Credentials are ignored by Git and are never embedded in the public repository.

Reproducible delivery

Docker supports local deployment; GitHub Actions runs Ruff and tests across supported Python versions.

Production path

A multi-user service should add PostGIS, caching, authentication, quotas, encrypted secrets, audit logs, and dedicated geocoding.

What this repository can demonstrate

Market orientation

Compare business density, category coverage, and contact completeness across cities or neighborhoods.

Data-engineering education

Teach adapter patterns, provenance, normalization, quality metrics, geospatial interfaces, and API design.

Responsible AI practice

Separate deterministic evidence from optional model interpretation and make credential ownership explicit.

Portfolio review

Show Python engineering, FastAPI, testing, CI, Docker, public documentation, scholarly metadata, and deployment planning.

Citation status: metadata is prepared in CITATION.cff, .zenodo.json, and codemeta.json. A version-specific DOI will be added after the first Zenodo-archived GitHub release.

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The map becomes intelligence only after the uncertainty remains visible.